RAG-BASED ENHANCEMENTS FOR CONTEXT-AWARE INTERFACES IN LARGE LANGUAGE MODELS

By

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Large Language Models (LLMs) have demonstrated strong capabilities in natural language understanding and generation, but they often produce incorrect or hallucinated responses when relying solely on internal model knowledge. Retrieval-Augmented Generation (RAG) addresses this limitation by grounding model outputs in externally retrieved context; however, the effectiveness of RAG systems is highly dependent on the quality of the retrieval component.

This thesis investigates the impact of retrieval strategies on the performance of RAG pipelines, with a particular focus on comparing single-stage dense retrieval against two-stage retrieval incorporating cross-encoder reranking. Using a controlled experimental setup on the CoQA dataset, pretrained bi-encoder and cross-encoder models are evaluated without task-specific fine-tuning. Retrieval effectiveness is measured using standard metrics including Hit@1 and Exact Match@1 (EM@1), enabling a clear assessment of how retrieval architecture influences grounding accuracy.

Experimental results demonstrate that two-stage retrieval with cross-encoder reranking consistently improves retrieval accuracy over dense retrieval alone, highlighting the importance of multi-stage retrieval pipelines in grounded question answering systems. These findings are further contextualized through an analysis of real-world RAG systems such as NotebookLM,

which employ structured retrieval, reranking, and strict source grounding to reduce hallucinations.

Based on the observed limitations of static retrieval pipelines, this thesis identifies Agentic RAG—where retrieval decisions are dynamically refined through iterative reasoning and tool selection—as a promising direction for future research. Overall, the study emphasizes retrieval quality as a critical factor in building reliable, context-aware, and trustworthy RAG-based language model systems.

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DEDICATION

To my parents, whose support, perseverance, and values have guided my academic and personal growth. Their encouragement has been fundamental to the completion of this work.

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TABLE OF CONTENTS

CHAPTER PAGE

ABSTRACT i

ACKNOWLEDGMENTS iii

DEDICATION iv

LIST OF TABLES vi

LIST OF FIGURES vii

CHAPTERS

CHAPTER 1 – INTRODUCTION 1

CHAPTER 2 – LITERATURE SURVEY 5

CHAPTER 3 – PROBLEM ANALYSIS 15

CHAPTER 4 – COMPARATIVE INSIGHTS 21

CHAPTER 5 – THEORITICAL FRAMEWORK 24

CHAPTER 6 – EXPERIMENTAL VALIDATION 34

CHAPTER 7 – STATE-OF-THE-ART 44

CHAPTER 8 – KEY FINDINGS AND CONCLUSION 52

REFERENCES 56

VITA 59

LIST OF TABLES

TABLE PAGE

Table 1 – Tabular Summary of Research works.....6

Table 2 – Comparative Strengths and Weaknesses across models.....22

Table 3 –Results (Performance Metrics) 41

LIST OF FIGURES

FIGURE PAGE

Figure 1 - Standard Retrieval-Augmented Generation (RAG) Pipeline Architecture.....27

Figure 2 -Theoretical Justification for the Framework.....32

Figure 3 - Graph RAG pipeline 46

Figure 4 - NotebookLM RAG pipeline architecture.....49

CHAPTER 1

INTRODUCTION

1.1 Background and Context

Large Language Models (LLMs) have rapidly advanced in their ability to understand and generate natural language, enabling applications across search systems, conversational agents, education, and decision-support tools. However, despite these improvements, LLMs still rely heavily on internal training data, which can lead to factual inconsistencies, outdated knowledge, and hallucinations when deployed in real-world settings.

Retrieval-Augmented Generation (RAG) addresses these limitations by grounding model outputs in externally retrieved documents, thereby improving factual accuracy and contextual relevance. As RAG systems evolve, research and industrial deployments increasingly emphasize the importance of retrieval quality, multi-stage retrieval pipelines, and controlled grounding mechanisms. More advanced architectures, including graph-based and agent-driven approaches, have been proposed to further enhance reasoning capabilities, though many practical systems continue to rely on structured but static retrieval pipelines. Understanding how retrieval strategies influence the reliability and interpretability of RAG-based systems is therefore critical for building trustworthy language model applications.

1.2 Problem Statement

Despite significant advancements in Large Language Models (LLMs), modern systems still struggle to produce reliable, contextually grounded, and factually consistent responses when relying solely on internal training data. Retrieval-Augmented Generation (RAG) provides partial improvements by integrating external knowledge, yet current RAG architectures suffer from shallow context fusion, inefficient or redundant retrievals, and limited multi-hop reasoning

capabilities. These limitations lead to hallucinations, poor interpretability, and weak reasoning performance in complex, real-world tasks. Therefore, there is a critical need for a structured, comparative examination of existing RAG and emerging Graph-RAG frameworks, along with the development of a theoretical foundation that improves context integration, reasoning depth, and reliability in LLM-driven applications.

1.3 Motivation for the Study

Although Retrieval-Augmented Generation improves factual grounding in Large Language Models, practical deployments reveal that retrieval quality remains a primary bottleneck for reliable performance. Even when generation models are highly capable, poorly retrieved context can lead to incorrect or misleading outputs.

The motivation for this study stems from the need to understand how retrieval architecture choices influence grounding accuracy in RAG systems, particularly in real-world settings where model retraining is not always feasible. By focusing on pretrained retrieval models and controlled retrieval pipelines, this work aims to provide empirical insight into the role of multi-stage retrieval and reranking in improving reliability.

Furthermore, the study is motivated by real-world RAG systems such as NotebookLM, which emphasize structured retrieval, reranking, and strict grounding. The observed limitations of static retrieval pipelines motivate future exploration of adaptive and agentic retrieval strategies.

1.4 Research Objectives

The primary objective of this thesis is to investigate how retrieval architecture choices affect the performance and reliability of Retrieval-Augmented Generation (RAG) systems. Rather than proposing new model architectures or knowledge representations, this work focuses

on a controlled empirical evaluation of retrieval strategies within RAG pipelines using pretrained models.

The specific research objectives of this study are:

1. To examine the limitations of standalone Large Language Models (LLMs) in producing factually grounded and contextually accurate responses when relying solely on internal training data.
2. To evaluate the effectiveness of single-stage dense retrieval (bi-encoder-based retrieval) in identifying relevant contextual passages for question answering tasks.
3. To analyze the impact of two-stage retrieval pipelines that incorporate cross-encoder reranking on retrieval accuracy and grounding quality.
4. To quantitatively assess retrieval performance using standard evaluation metrics, including Hit@1 and Exact Match@1 (EM@1), in a controlled experimental setting.
5. To compare retrieval outcomes across different retrieval strategies while keeping the generation component fixed, thereby isolating the role of retrieval architecture in RAG performance.
6. To situate the experimental findings within the context of real-world RAG systems, such as NotebookLM, and identify architectural patterns that emphasize grounding, reranking, and retrieval precision.
7. To motivate Agentic RAG as a future research direction by identifying limitations of static retrieval pipelines observed through empirical analysis.

1.5 Scope and Limitations

The scope of this thesis is limited to the evaluation of retrieval strategies within text-based Retrieval-Augmented Generation (RAG) systems. The study focuses specifically on

comparing single-stage dense retrieval with two-stage retrieval pipelines that incorporate cross-encoder reranking. All experiments are conducted using pretrained retrieval models without task-specific fine-tuning, allowing for a clear comparison of architectural differences rather than training effects. The experimental evaluation is performed on a subset of the CoQA dataset and uses retrieval-focused metrics, including Hit@1 and Exact Match@1 (EM@1), to measure grounding effectiveness. The generation component of the RAG pipeline is not modified or optimized, as the primary emphasis of this work is on retrieval quality rather than language generation performance. This thesis does not propose or implement new retrieval algorithms, graph-based knowledge representations, or agentic control mechanisms. While concepts such as Graph-based RAG and Agentic RAG are discussed to contextualize the work within the broader research landscape, they are not experimentally evaluated. Additionally, the study does not address multimodal RAG systems, domain-specific retrieval scenarios, large-scale benchmarking, or deployment-level considerations. These scope boundaries ensure that the thesis maintains a focused and defensible analysis of retrieval architecture choices in RAG systems, while clearly identifying directions for future research beyond the current experimental framework.

CHAPTER 2

LITERATURE SURVEY

2.1 Overview of Retrieval-Augmented Generation Research

Retrieval-Augmented Generation (RAG) has emerged as a foundational approach for improving the factual grounding, reliability, and context-awareness of Large Language Models (LLMs). Early transformer-based LLMs demonstrated impressive generative fluency but struggled with hallucinations, outdated knowledge, and limited reasoning when relying solely on internal parameters. RAG addresses these limitations by coupling an external retriever with a generator, enabling models to incorporate relevant documents during inference and produce responses grounded in real evidence.

The retriever–generator paradigm sparked rapid innovation across both retrieval strategies and fusion techniques. Dense embedding–based retrievers such as DPR improved semantic matching, while models like FiD and ColBERT introduced more sophisticated multi-document fusion mechanisms. Subsequent works explored step-wise reasoning integration (e.g., Chain-of-Thought RAG) and retrieval-guided reasoning pipelines. Despite these advancements, traditional RAG architectures remained primarily text-chunk–driven, limiting their ability to capture relational knowledge and perform multi-hop reasoning.

This led to the emergence of **Graph-based RAG** approaches, which structure retrieved information as entities and relationships rather than isolated text segments. GraphRAG, Enhanced GraphRAG, and similar systems highlight a paradigm shift from retrieval-focused to reasoning-aware RAG, enabling models to infer through connected evidence, reduce redundancy, and produce more interpretable outputs.

Overall, the evolution of RAG research reflects a transition from simple document

retrieval to increasingly structured, relational, and reasoning-oriented augmentation. This subsection provides the foundation for analyzing six representative papers, whose methodologies, contributions, and limitations are summarized in the following sections.

2.2 Overview of Selected Research Works

This literature survey examines six influential research works that represent key developments in Retrieval-Augmented Generation (RAG), context-aware reasoning, and graph-structured retrieval for Large Language Models (LLMs). These papers were selected because they collectively highlight the evolution from traditional retrieval mechanisms toward more structured, reasoning-oriented, and context-integrated frameworks.

The selected works span three major categories:

1. **Foundational RAG architectures** that focus on integrating external retrieval with generation.
2. **Decoder-fusion and late-fusion models** enabling multi-document understanding.
3. **Graph-based retrieval and reasoning frameworks** that push LLMs toward relational, multi-hop, and interpretable reasoning.

Together, these six papers provide a representative view of the current landscape, enabling comparisons across retrieval precision, reasoning depth, context-fusion strategies, and practical limitations. The sections that follow analyze each paper's methodological contributions and limitations, synthesize comparative insights, and highlight the gaps motivating a unified theoretical framework.

2.3 Tabular Summary of Research Works.

Paper / Framework	Main Contribution	Methodology	Limitations
RAG (Lewis et al., 2020)	Introduced retrieval-augmented	Dense retriever + seq-to-seq generator;	Retrieval is static; shallow context

	generation; improved factual grounding of LLMs	concatenates retrieved documents with prompt	fusion; weak multi-hop reasoning
Fusion-in-Decoder (FiD)	Enhanced multi-document fusion in encoder-decoder models	Decoder-level fusion of retrieved passages; processes each document independently before decoding	High computational cost; still limited relational reasoning
ColBERT	Efficient late-interaction model for scalable retrieval	BERT-based token scoring with late interaction; vector-based retrieval	Focuses on retrieval, not reasoning; requires large index storage
CoT-RAG	Added chain-of-thought reasoning on top of retrieval	Retrieves evidence → generates reasoning steps using CoT prompts	Expensive multi-step generation; reasoning not fully grounded
Graph RAG	Introduced graph-structured retrieval for multi-hop reasoning	Converts corpus → entities/relations → knowledge graph; retrieval through graph traversals	Requires graph construction; domain-specific; expensive to scale
Enhanced Graph RAG	Improved interpretability and context fusion using graph clustering	Clusters subgraphs, aggregates evidence, and enables relational reasoning during generation	High computational overhead; limited real-world evaluations

Table 1: Tabular Summary of Research Works.

2.4 Review of Key RAG and Graph-RAG Papers

This section provides a detailed review of the six foundational papers that shape the evolution of Retrieval-Augmented Generation (RAG) and its extensions. These works demonstrate how retrieval mechanisms, document fusion strategies, and structured knowledge representations influence factual accuracy, reasoning depth, and contextual grounding in Large Language Models.

2.4.1 RAG: Retrieval-Augmented Generation (Lewis et al., 2020)

Lewis et al. introduced RAG as a hybrid architecture combining dense retrieval with seq-

to-seq generation. The retriever, based on DPR embeddings, fetches top-k relevant text chunks, while a generative model (typically BART) synthesizes responses using both the prompt and retrieved evidence. The model improved factual correctness but had shallow context integration, as retrieved passages were concatenated linearly without reasoning structure. This foundational work established retrieval as a crucial mechanism for reducing hallucinations.

2.4.2 FiD: Fusion-in-Decoder (Izacard & Grave, 2021)

FiD enhanced multi-document comprehension by shifting the fusion process entirely into the decoder. Instead of concatenating passages as a single input, FiD encodes each document independently and merges them during decoding. This architecture increased the model's ability to handle long-context retrieval and multi-evidence reasoning. However, FiD demands significantly more memory and computational resources due to encoding multiple documents separately.

2.4.3 ColBERT: Contextualized Late Interaction (Khattab & Zaharia, 2020)

ColBERT emerged as a breakthrough in high-precision retrieval. Using late interaction mechanisms and contextualized token embeddings, it delivered retrieval that is both semantically fine-grained and computationally efficient. Its strengths lie in high recall and relevance, which directly benefit RAG pipelines. However, ColBERT is a retriever-only method and does not specify how retrieved context should be fused during generation, limiting its standalone reasoning capabilities.

2.4.4 CoT-RAG: Chain-of-Thought-Augmented Retrieval (Press et al., 2022)

CoT-RAG integrates structured reasoning into the retrieval process. Instead of retrieving text upfront, the LLM generates step-wise Chain-of-Thought (CoT) reasoning traces that inform dynamic retrieval queries. This leads to better grounding and transparency but remains sensitive

to errors in intermediate CoT steps. The method improved interpretability but introduced instability when CoT sequences diverged from the intended task.

2.4.5 GraphRAG: Graph-Structured Retrieval (2023)

GraphRAG introduced a graph-based paradigm where text is converted into knowledge graphs containing entities, relations, and local subgraphs. Instead of retrieving isolated chunks, the system retrieves graph-connected nodes, enabling multi-hop reasoning across related concepts. This yields superior contextual coherence and interpretability. However, constructing and maintaining high-quality graphs is computationally expensive, and performance heavily depends on the graph's correctness.

2.4.6 Enhanced GraphRAG (2024)

Enhanced GraphRAG extends GraphRAG by incorporating relationship-aware fusion and cluster-based aggregation. Retrieved subgraphs undergo graph-aware CoT reasoning, enabling stronger multi-step logical inference. While this yields high reasoning depth and better factual consistency, large knowledge graphs, and domain transferability. It also presents higher computational overhead during graph generation and clustering.

2.5 Methodological Trends Across RAG Architectures

Recent advancements in Retrieval-Augmented Generation (RAG) systems reveal several methodological trends that shape how modern LLMs retrieve, organize, and integrate external knowledge. While early architectures primarily emphasized retrieval precision, contemporary systems increasingly incorporate structured representations, reasoning-aware fusion, and multi-hop retrieval mechanisms to address the limitations of traditional, linear pipelines. The following subsections summarize the dominant methodological patterns observed across the six RAG and Graph-RAG papers reviewed.

2.5.1 Shift From Linear Retrieval to Multi-Stage Hybrid Retrieval

Traditional RAG architectures (e.g., Lewis et al., 2020) use a single-step dense or sparse retriever such as DPR or BM25. More recent models incorporate **hybrid retrieval pipelines**, combining:

- **Dense retrieval** (semantic embeddings)
- **Sparse retrieval** (keyword-level precision)
- **Entity or relationship-based retrieval** (GraphRAG, Enhanced GraphRAG)

This hybrid strategy reduces redundancy and improves the diversity of retrieved evidence, which is a critical step in multi-document reasoning tasks.

2.5.2 Increasing Emphasis on Structured Knowledge Representations

A major trend across contemporary RAG approaches is the shift from treating retrieved text as flat, unstructured information toward **structured knowledge formats**, including:

- Entity–relation graphs
- Local subgraphs or clusters
- Topic-level or document-level hierarchies

GraphRAG, in particular, converts retrieved documents into **node–edge representations**, enabling models to exploit relational pathways (A → B → C) during reasoning. This trend significantly enhances multi-hop inference, contextual grounding, and interpretability.

2.5.3 Evolution From Simple Concatenation to Advanced Context Fusion

Early RAG pipelines simply concatenated retrieved passages with the query and passed them into the generator. This approach offered limited reasoning depth and often overwhelmed the generator with redundant or irrelevant text.

Newer architectures propose more sophisticated fusion methods:

- **Decoder-level fusion** (FiD) for multi-document comprehension
- **Step-wise reasoning fusion** (CoT-RAG) for explanation-aware generation
- **Graph-guided fusion** (GraphRAG) using relationship-aware aggregation

These methods improve coherence, factual grounding, and the model's ability to use retrieved evidence effectively.

2.5.4 Rise of Explicit Reasoning Strategies

Another trend is the integration of explicit reasoning to enhance factual consistency. Approaches include:

- **Chain-of-Thought reasoning** embedded into the generation process
- **Graph expansion-based reasoning** for relational inference
- **Structured rationale generation** in Explainable RAG systems

These methods allow models to justify their outputs, reduce hallucinations, and better align reasoning chains with retrieved evidence.

2.5.5 Incorporation of Adaptive Retrieval Techniques

Static retrieval—where the retriever only runs once at the beginning of the pipeline—has proven insufficient for complex tasks requiring dynamic context. As a result, adaptive retrieval strategies have emerged, such as:

- **ReAtt (Retrieval Attention)**: retrieves mid-generation
- **FLARE**: retrieves when uncertainty is detected
- **Iterative graph expansion** in GraphRAG

These adaptive strategies enable more context-aware responses and reduce over-reliance on initial retrieval.

2.5.6 Increasing Focus on Efficiency and Scalability

As RAG systems become more complex, computational efficiency becomes a challenge, especially for graph-based models. Current research trends include:

- Reducing redundant retrieval using clustering
- Using lightweight graph construction
- Selective expansion of subgraphs
- Optimizing vector store queries

These directions aim to make RAG systems scalable for real-world deployment.

2.6 Identified Limitations and Research Gaps

1. Inadequate Context Fusion Mechanisms

Most traditional RAG systems—including RAG (Lewis et al.), FiD, and ColBERT—combine retrieved documents using shallow concatenation or decoder-level fusion. While effective for simple retrieval tasks, these methods fail to integrate relational knowledge or multi-hop dependencies across documents. This results in **fragmented evidence integration** and **incomplete reasoning**, especially in tasks requiring deep understanding of entity relationships or temporal ordering.

2. Limited Multi-Hop and Relational Reasoning

Even advanced models that incorporate step-wise reasoning (e.g., CoT-RAG) still rely heavily on the LLM's latent reasoning abilities. Graph-RAG frameworks improve structured reasoning but remain **domain-dependent**, requiring pre-built or curated knowledge graphs. These approaches struggle with:

- dynamic graph construction,
- noisy or ambiguous retrieved evidence,
- maintaining consistency across multi-step reasoning chains.

Thus, **robust and domain-general multi-hop reasoning** is still unresolved.

3. Retrieval Redundancy and Precision Issues

Dense retrieval frameworks often retrieve overlapping or semantically redundant passages, limiting diversity. Conversely, sparse retrieval methods miss semantically similar but lexically different contexts. Hybrid retrieval helps, but no system fully addresses:

- redundant chunk retrieval,
- missing fine-grained relational evidence,
- balancing retrieval diversity and precision.

This leads to **suboptimal grounding** and reduces model efficiency.

4. Computational Cost and Scalability Constraints

Graph-enhanced RAG systems, particularly Enhanced GraphRAG, require intensive computation for graph extraction, community detection, and multi-hop traversal. These approaches are often too slow or resource-heavy for real-time deployments. Large-scale graph construction is especially problematic for dynamic or evolving domains, making **practical scalability a major challenge**.

5. Poor Interpretability of Reasoning Paths

Most RAG models provide output without explaining **how** retrieved documents influenced the final response. While GraphRAG introduces interpretable reasoning paths, traditional RAG models lack structured attribution mechanisms. This hinders trust, especially in high-stakes applications such as healthcare or finance.

6. Lack of Unified Evaluation Metrics

Current evaluations measure factuality, retrieval accuracy, or response quality independently, but **no standardized metric evaluates context fusion or reasoning depth**. The

absence of unified benchmarks makes cross-paper comparison difficult and limits reproducibility.

7. Limited Generalization Across Domains

Graph-based and hybrid retrieval approaches often rely on domain-specific heuristics and graph structures. As a result, their performance does not transfer well to:

- open-domain tasks,
- highly specialized domains (legal, medical),

This creates a gap for developing **general-purpose context-aware reasoning frameworks**.

CHAPTER 3

PROBLEM ANALYSIS

3.1 Inadequate Retrieval Precision in Existing RAG Systems

Retrieval-Augmented Generation was introduced to mitigate hallucinations by grounding LLM responses in external evidence. However, existing retrieval pipelines still struggle to consistently fetch *high-quality, diverse, and contextually aligned* information. This creates several challenges that directly limit reasoning quality:

1. Retrieval Redundancy

Dense retrievers (e.g., DPR, ColBERT) often retrieve multiple passages that repeat the same information, reducing informational diversity. This leads to inefficient context fusion and limits the model's ability to reason across varied evidence sources.

2. Shallow Semantic Matching

Most retrievers prioritize lexical or embedding similarity rather than conceptual or relational relevance. As a result, retrieved documents may be topically related but *not* contextually meaningful for the specific question or reasoning step.

3. Missing Multi-Hop Evidence

Many reasoning tasks require multi-step evidence chains (A → B → C). Traditional RAG systems rarely retrieve sequential, relationally-linked information, causing gaps in understanding and preventing coherent multi-hop reasoning.

4. Domain Shift & Poor Generalization

Retrievers trained on general-domain corpora often underperform on specialized datasets (e.g., finance, science) due to vocabulary mismatch and distributional gaps. As a result, retrieval noise increases, and grounding reliability drops.

5. Lack of Structural Awareness

Text chunks retrieved as isolated units lack the relational structure needed for reasoning. This is a major bottleneck where Graph RAG systems begin to outperform traditional RAG.

3.2 Fragmented Context Fusion and Weak Integration Mechanisms

A major challenge in current RAG architectures is their inability to **integrate retrieved information into a coherent and contextually aligned representation**. While retrieval modules in systems like RAG, FiD, and ColBERT successfully fetch relevant documents, the **fusion mechanisms** responsible for combining retrieved evidence with generation are often shallow and structurally weak.

Many first-generation RAG models rely on **simple concatenation** of retrieved text chunks before decoding. Although this allows the model to condition on external knowledge, it frequently leads to **redundancy**, **context dilution**, and difficulty in distinguishing between primary and supporting evidence. This linear fusion approach lacks awareness of **semantic relationships**, **entity dependencies**, or **cross-document coherence**, causing the model to treat all retrieved passages as independent fragments.

Even advanced fusion approaches, such as FiD's decoder-level attention, improve multi-document absorption but still operate on **token-level interactions** without explicitly modeling the relationships between retrieved concepts. As a result, the generator can miss deeper semantic structures—such as causal relationships, multi-hop reasoning paths, or hierarchical concepts—that are essential for accurate and logically grounded responses.

Graph-based RAG models attempt to address this issue through entity-relationship graphs and local neighborhood clustering. However, these methods remain limited by **expensive graph construction**, **domain dependence**, and **inconsistent scalability** across large corpora.

Moreover, the transition from retrieved text to graph nodes introduces abstraction gaps, making real-time reasoning challenging.

Overall, the weak integration between retrieved evidence and LLM generation leads to **fragmented understanding**, **loss of contextual relevance**, and increased likelihood of **hallucinations**. These limitations highlight the need for structured and relationship-aware fusion mechanisms capable of preserving semantic continuity and supporting deeper reasoning processes.

3.3 Limited Multi-Hop, Logical, and Relational Reasoning

While Retrieval-Augmented Generation (RAG) architectures significantly enhance a model's access to external information, their ability to **reason** over that information remains fundamentally limited. Traditional RAG pipelines excel at *retrieving* relevant content, but they fall short when tasks require **multi-step inference, causal chaining, relational understanding, or reasoning across multiple retrieved segments**. This limitation becomes more evident as tasks shift from simple fact recall to complex analytical reasoning.

A key challenge lies in the **linear and unstructured nature of retrieved text**. Standard RAG models concatenate retrieved passages and pass them directly to a generator (e.g., FiD, DPR+Generator). While this improves factual grounding, it does not guarantee that the model can **connect** information across passages. As a result, LLMs frequently perform *shallow pattern matching* instead of deep reasoning.

Furthermore, traditional retrieval systems are inherently **local and myopic**; they retrieve chunks that seem textually relevant, but not necessarily relationally connected. This prevents models from performing tasks such as:

- inferring **A → B → C** causal chains,

- synthesizing information scattered across documents,
- following reasoning paths spanning multiple entities or concepts,
- resolving long-range dependencies or multi-document relationships.

Even enhanced diffusion-style models like FiD improve fusion but do not fundamentally address the **reasoning gap** — they fuse more text, but do not reason more deeply.
More recent approaches, such as **CoT-RAG**, attempt to induce chain-of-thought style reasoning by retrieving evidence iteratively. However, these methods remain limited because they rely on implicit LLM reasoning without structured support. They improve explanation generation but still suffer when the underlying retrieval lacks relational structure.

Graph-based approaches (e.g., GraphRAG, Enhanced GraphRAG) partially address this by transforming retrieved content into **entity–relationship graphs**, enabling the model to navigate structured connections. Despite this, challenges remain: graph construction is often brittle, domain-dependent, and computationally expensive, and many graphs fail to capture the full semantic richness of unstructured text.

Overall, today's RAG systems continue to rely heavily on **LLM inference heuristics**, which are not reliable enough for high-stakes reasoning. Without explicit structures supporting multi-hop inference, current systems exhibit:

- failure to maintain reasoning coherence,
- hallucinated or unsupported conclusions,
- and an inability to explain their reasoning paths.

These limitations highlight the critical need for **reasoning-aware RAG architectures** that can combine retrieval with structured inference mechanisms, enabling more interpretable, accurate, and robust multi-step reasoning.

3.4 Scalability Constraints and Computational Inefficiencies

Although RAG and Graph RAG architectures strengthen factuality and reasoning, they introduce substantial computational overhead that limits their scalability in real-world deployments. Traditional RAG models rely on dense retrieval mechanisms (e.g., DPR, ColBERT), which require large vector databases and high-dimensional embeddings. As the corpus size grows, retrieval latency increases significantly, forcing systems to trade off between retrieval precision and system responsiveness. This makes real-time applications—such as conversational agents or continuous monitoring systems—difficult to maintain at scale. Graph-based extensions, while more powerful in reasoning, further intensify computational demands. Constructing knowledge graphs from large document collections requires expensive entity extraction, relationship prediction, and graph clustering. Inference on these graphs introduces additional complexity: multi-hop reasoning often involves traversing thousands of interconnected nodes, which increases memory usage and slows generation. Many existing Graph RAG systems also rely on repeated graph traversals for each query, making them difficult to deploy on resource-constrained environments.

Hybrid retrieval pipelines that combine dense embedding search with graph-guided reranking also face performance bottlenecks. While they improve retrieval quality, they require multiple passes over the data, increasing both computational and financial costs. As a result, most current RAG systems struggle to scale beyond moderate-sized corpora or require infrastructure (GPU clusters, high-speed vector stores, distributed indexing) that is impractical for smaller research teams and organizations.

Overall, scalability remains a central challenge: achieving high-quality retrieval and reasoning while maintaining efficiency, low latency, and reasonable resource consumption. This

limitation creates a barrier to deploying advanced RAG models in real-world, high-volume, or low-resource contexts.

3.5 Limited Interpretability and Absence of Transparent Reasoning Paths

A critical problem observed across existing RAG and Graph-RAG systems is the lack of **clear, interpretable reasoning traces** that explain *why* a model retrieved specific documents and *how* those documents influenced the final answer. While retrieval-augmented systems significantly improve factual grounding, they still behave largely as **black boxes**, offering limited visibility into the internal steps that lead to a generated response.

The absence of transparency creates several challenges:

- **Difficulty validating correctness:** Users cannot confirm whether the model relied on appropriate, relevant evidence versus superficial keyword overlaps.
- **Lack of trust in high-stakes domains:** Fields such as healthcare, finance, education, and policy require models to justify outputs with interpretable rationale.
- **Hard to debug model failures:** When a system hallucinates or misinterprets context, developers lack insights into which retrieval step or fusion mechanism went wrong.
- **Barrier to regulatory compliance:** Many evolving AI safety and governance frameworks require explainability, which current RAG systems do not fully satisfy.

This limitation ultimately restricts the deployment of RAG-based systems in environments where decisions must be **auditable, accountable, and traceable**. Therefore, enhancing interpretability—through reasoning-aware retrieval, structured knowledge tracing, or graph-based explanation layers—is essential for advancing reliable and transparent context-aware LLMs.

CHAPTER 4

COMPARATIVE INSIGHTS

4.1 Evolution of Retrieval Approaches

- **Traditional RAG** relies on dense embeddings (DPR, FAISS), offering fast retrieval but limited relational awareness.
- **FiD and ColBERT** improve evidence coverage using multi-document fusion or late interaction scoring.
- **CoT-RAG** adds chain-of-thought guidance to retrieval, improving interpretability.
- **GraphRAG and Enhanced GraphRAG** incorporate knowledge graph construction, enabling multi-hop evidence tracing and reducing redundancy.

Across these systems, retrieval quality increasingly depends on **how well the model assembles diverse, complementary evidence**, not just how many documents it retrieves.

4.2 Trends in Context Integration and Fusion

A major differentiator among RAG-based systems is the **fusion layer**, where retrieved evidence is combined with the LLM:

- **RAG** uses simple concatenation → often leads to repetition and shallow grounding.
- **FiD** improves fusion by encoding documents separately and integrating them at the decoder.
- **CoT-RAG** incorporates explicit reasoning steps, offering structure on *how* the context is fused.
- **GraphRAG** uses graph-guided aggregation, creating relational clusters before passing them to the generator.

The overall trend is a movement toward **structured and selective context integration**, reducing noise and strengthening internal reasoning pathways.

4.3 Variation in Reasoning Capabilities

Reasoning depth varies widely across the surveyed systems:

- **RAG** → minimal reasoning, relies on LLM's implicit ability.
- **FiD** → supports multi-document reasoning but without explicit logical structure.
- **CoT-RAG** → introduces stepwise reasoning that improves transparency and reliability.
- **GraphRAG** → supports multi-hop and relational reasoning via graph traversal.

This demonstrates a transition from **implicit reasoning** to **explicit, guided reasoning**, reflecting an emerging consensus that reasoning should be treated as a core architectural component rather than a side effect of generation.

4.4 Comparative Strengths and Weaknesses Across Models

Model	Strengths	Weaknesses
RAG	Fast, simple to implement; strong factuality boost	Shallow context fusion; limited reasoning
FiD	Strong multi-document integration; robust retrieval coverage	High compute cost; implicit reasoning only
ColBERT	Late interaction improves fine-grained retrieval	Not optimized for reasoning or fusion
CoT-RAG	Transparent stepwise reasoning; better grounding	Reasoning is still linear; may overfit irrelevant steps
GraphRAG	Multi-hop reasoning; graph explains relationships	Graph construction overhead; domain dependency
Enhanced GraphRAG	Best relational reasoning; structured fusion pipeline	Expensive, complex, less scalable

Table 2: Comparative Strengths and Weaknesses Across Models

This comparison confirms that **no single architecture fully solves retrieval, fusion, and reasoning simultaneously**, motivating the need for more unified frameworks.

4.5 Key Takeaways From Comparative Analysis

- Retrieval is becoming **multi-strategy**, incorporating dense, sparse, and graph-informed

methods.

- Fusion is evolving from concatenation to **structured, relationship-aware integration**.
- Reasoning shifts from being implicit to **explicit and guided**, improving interpretability.
- Graph-based approaches offer the strongest reasoning performance but introduce scalability and cost challenges.
- Existing systems lack a unified pipeline that optimizes **retrieval precision, context fusion, and reasoning depth together**.

These insights lay the foundation for the next chapter, where your **proposed conceptual framework** builds on these observed patterns to address major gaps in current architectures.

CHAPTER 5

THEORETICAL FRAMEWORK

5.1 Conceptual Foundations of Context-Aware RAG

Retrieval-Augmented Generation (RAG) emerged as a foundational approach to reduce hallucinations in Large Language Models by grounding generation in retrieved external evidence. However, traditional RAG pipelines assume that simple retrieval + concatenation is sufficient for complex reasoning tasks. As LLM applications expanded into domains requiring multi-step inference, temporal coherence, relational understanding, and deeper contextual alignment, it became clear that the classical RAG formulation is fundamentally limited.

A context-aware RAG framework is built upon the idea that *retrieval alone is not enough*—the model must also understand *how pieces of retrieved information relate to one another* and *how they should be fused* before response generation. The shift from retrieval-centric architectures to reasoning-centric architectures represents the conceptual evolution motivating this framework.

The conceptual foundation of a context-aware RAG system rests on three principles:

1. Structured Retrieval Over Flat Document Fetching

Traditional k-nearest neighbor retrieval retrieves text chunks independently, ignoring semantic or relational structure.

Context-aware RAG assumes retrieval should preserve:

- entity relationships
- concept-level dependencies
- multi-hop connections across sources
- evidence hierarchies (primary vs. supporting)

This structure allows the model to reason *through* content rather than merely *over* content.

2. Intermediate Knowledge Organization Before Fusion

Instead of feeding raw text blocks to the generator, context-aware RAG processes retrieved documents to produce:

- entity graphs
- relationship maps
- hierarchical clusters
- logical or thematic groupings

This intermediate organization mirrors human comprehension strategies, enabling more coherent fusion and reducing redundancy.

3. Reasoning-Guided Fusion Rather than Linear Concatenation

Context-aware RAG employs reasoning signals—implicit (LLM-internal) or explicit (chain-of-thought, graph traversal)—to determine:

- which information is most relevant
- how pieces of evidence support one another
- how inconsistencies should be resolved
- what reasoning path leads to the final answer

This results in better factual consistency, richer contextual grounding, and more interpretable outputs.

5.2 Architecture of the Proposed Context Fusion Framework

The proposed Context Fusion Framework introduces structured, multi-layered architecture designed to improve factual grounding, contextual coherence, and reasoning quality within Retrieval-Augmented Generation (RAG) systems. Unlike conventional RAG pipelines

that rely on shallow concatenation of retrieved documents, this framework restructures information into more meaningful representations before integrating it into the generation process. The architecture is organized into **four sequential layers**, each addressing a specific weakness identified in existing RAG and Graph RAG models:

1. Hybrid Retrieval Layer

This layer combines the strengths of dense retrieval, sparse retrieval, and lightweight entity extraction to collect diverse and non-redundant evidence. Dense retrievers identify semantically similar passages, while sparse retrievers preserve keyword specificity, ensuring balanced recall and precision. Entity and relationship extraction components capture key terms and their connections early in the pipeline, enabling richer downstream structuring. This hybrid approach overcomes the limitations of single-method retrieval strategies, which often produce redundant or context-poor evidence.

2. Knowledge Structuring Layer

Once retrieved, textual evidence is transformed into **structured representations** such as entity nodes, relationship edges, and coherent subgraphs. This step reduces semantic redundancy, clusters related concepts, and surfaces implicit relationships hidden across documents. By converting unstructured text into structured knowledge units, the framework allows subsequent layers to perform more targeted reasoning and supports multi-hop relational inferences similar to those in Graph RAG systems, but with lower computational overhead.

3. Context Fusion Layer

Traditional RAG models simply concatenate retrieved documents, resulting in weak integration and limited interpretability. This layer introduces **relationship-aware fusion**, **graph-guided aggregation**, and **stepwise reasoning integration**. Instead of merging evidence linearly,

the model aligns retrieved information using shared entities, relationship paths, and concept clusters. The fused context is then organized into a reasoning-ready representation that prioritizes relevance, eliminates duplicates, and forms explicit rationale chains for the generation process.

4. Reasoning-Guided Generation Layer

The final layer integrates the structured, fused context into the language model using reasoning-aligned prompts or chain-of-thought–style scaffolding. Rather than generating responses directly from raw retrievals, the LLM leverages structured evidence paths, enabling more factual, coherent, and interpretable outputs. This layer supports rationale traceability, allowing the model to justify its responses using the structured knowledge generated in earlier stages. As a result, the system provides responses that are not only contextually grounded but also easier to audit and verify.

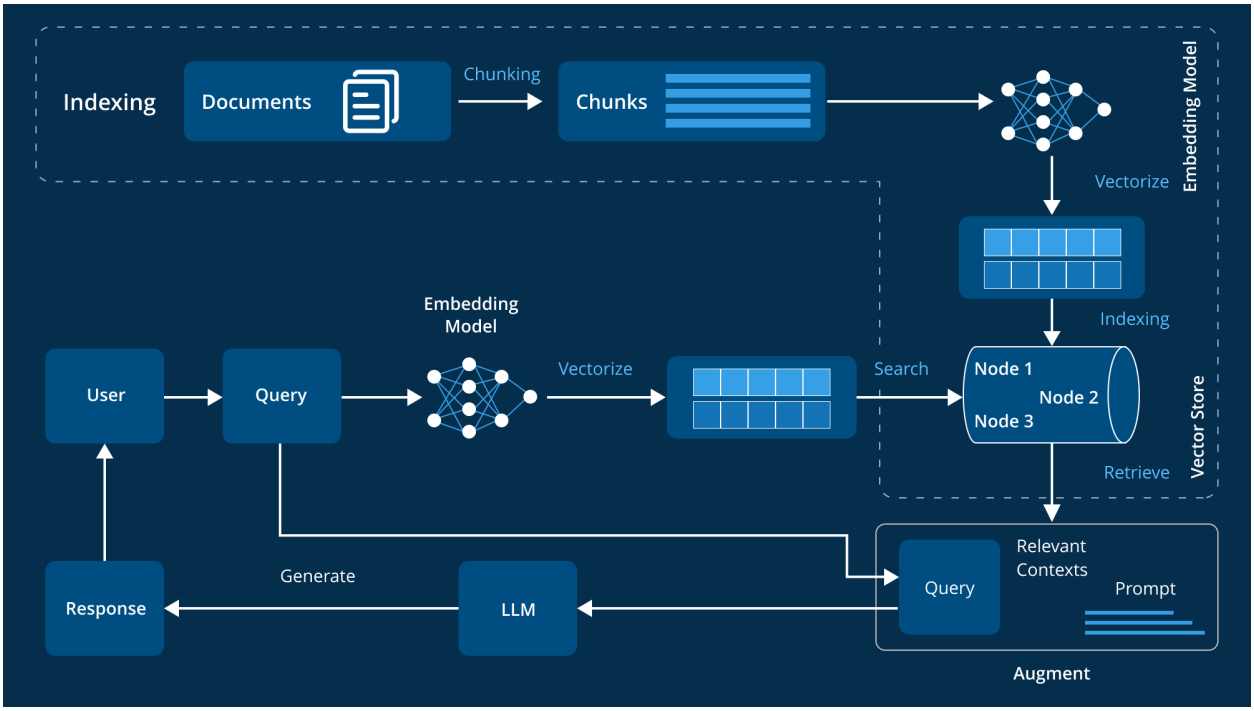

Fig 1: Standard Retrieval-Augmented Generation (RAG) Pipeline Architecture

5.3 Functional Components and Process Flow

The proposed context-aware RAG framework operates through a sequence of tightly integrated functional components, each responsible for transforming raw user input into grounded, context-rich, and reasoning-aligned responses. This section outlines the end-to-end flow of information through the system, highlighting how retrieval, structuring, fusion, and generation interact to improve factuality and interpretability in Large Language Models.

5.3.1 User Query Intake

The process begins when the user submits a query—either a natural-language question or a task description. Unlike standard RAG systems that immediately embed the query, this framework routes the query through a preprocessing stage that performs normalization, entity detection, and ambiguity checks. This ensures that subsequent retrieval and reasoning steps operate on a well-defined representation of the user's intent.

5.3.2 Hybrid Retrieval Mechanism

After preprocessing, the system triggers the hybrid retrieval layer, which performs two parallel operations:

- **Dense Retrieval:** Uses vector embeddings to locate semantically similar passages.
- **Sparse/Keyword Retrieval:** Captures exact matches, named entities, and symbolic cues often missed by dense retrievers.

These two retrieval modes complement each other, reducing blind spots and improving the overall recall of relevant information.

5.3.3 Knowledge Structuring and Graph Construction

Retrieved text is then converted into a structured representation. This step includes:

- Extracting **entity nodes**

- Identifying **relationship edges**
- Forming **local clusters or subgraphs**
- Merging redundant or overlapping information

This structured layer positions the system to perform relational reasoning rather than operate on isolated text fragments. It is particularly important for multi-hop inference, cause–effect reasoning, and concept disambiguation.

5.3.4 Context Fusion Layer

In this step, the system fuses retrieved and structured information into a coherent context. Instead of simple concatenation, the framework applies:

- **Relationship-aware aggregation**
- **Graph-guided fusion**, where information flows along entity–relation links
- **CoT-style structured reasoning**, which decomposes the problem into interpretable steps

This layered reasoning process ensures that the generated context reflects logical coherence and preserves evidence chains across documents.

5.3.5 Prompt Construction and Augmentation

The fused, structured context is then embedded into a prompt template. The model supplements the user query with:

- Extracted knowledge clusters
- Relevant evidence paths
- Step-wise reasoning indicators
- Condensed summaries of retrieved material

This augmented prompt directs the LLM to operate on curated, well-integrated context rather than raw retrieved text.

5.3.6 Reasoning-Guided Generation

Finally, the LLM uses the enriched prompt to produce the output. Because the input is structured and relationally grounded, the generated responses exhibit:

- Higher factual accuracy
- Reduced hallucination tendencies
- Better alignment with retrieved evidence
- Clearer reasoning traces

The system can also output rationales or evidence paths, improving interpretability in high-stakes applications.

5.4 Theoretical Justification for the Framework

The proposed Context Fusion Framework is theoretically grounded in the limitations identified in earlier RAG architectures and the cognitive principles underlying multi-step reasoning. Traditional RAG pipelines rely heavily on linear retrieval and concatenation-based fusion, which fail to capture structural relationships, contextual dependencies, or multi-hop reasoning patterns. By contrast, the layered design of the proposed framework aligns with well-established theories in information retrieval, graph-based knowledge modeling, and structured reasoning.

The **Hybrid Retrieval Layer** is justified through retrieval theory, where combining sparse and dense retrieval methods improves recall and reduces semantic blind spots. Empirical studies in hybrid retrievers show that lexical matchers capture surface-level relevance, while embedding models capture semantic similarity—together improving retrieval precision and diversity.

The **Knowledge Structuring Layer** draws from graph theory and cognitive knowledge

representation models, which show that humans encode and reason over information using entities and relationships rather than isolated text. Converting retrieved chunks into structured graphs allows the system to model multi-hop dependencies, reduce redundancy, and explicitly represent relational pathways. This provides a representational advantage over unstructured concatenation used in most classical RAG models.

The **Context Fusion Layer** is theoretically supported by research in multi-document summarization and evidence aggregation. Graph-guided aggregation and CoT-style reasoning enable the model to integrate evidence coherently, avoiding the context fragmentation frequently observed in decoder-level or naive concatenation-based fusion. This structured fusion directly addresses the gap of inconsistent or shallow reasoning in prior RAG designs.

Finally, the **Reasoning-Guided Generation Layer** builds on LLM reasoning theory, which suggests that structured prompts and explicit reasoning pathways reduce hallucination rates and improve factual grounding. By allowing the model to operate over a structured context graph, the framework aligns generation with relational evidence rather than relying solely on probabilistic token prediction.

Overall, the theoretical justification emphasizes that structured knowledge, layered retrieval, and reasoning-aware fusion collectively create a more scalable, interpretable, and reasoning-capable system. This layered architecture systematically resolves the limitations identified in existing RAG and Graph RAG approaches, providing a solid theoretical foundation for improved context-aware LLM performance.

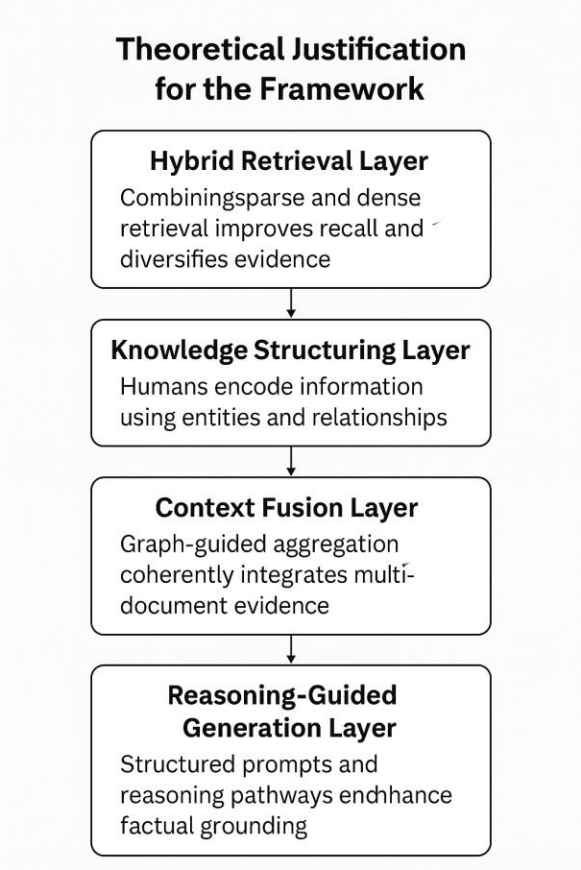

Figure 2: Theoretical Justification for the Framework

5.5 Expected Advantages and Impact

The proposed Context Fusion Framework offers several theoretical and practical advantages that directly address the limitations observed in existing RAG and Graph-RAG architectures. By combining hybrid retrieval, structured knowledge representation, graph-guided fusion, and reasoning-aligned generation, the framework enhances the reliability, depth, and interpretability of LLM outputs.

First, the framework supports **stronger factual grounding**, as retrieved evidence is not only retrieved more accurately but also organized into structured forms that reduce redundancy and highlight meaningful relationships. This decreases the likelihood of LLM hallucinations and ensures that answers remain tied to verifiable sources.

Second, the use of structured representations and multi-hop relational reasoning enables **more coherent and context-rich outputs**, especially for tasks requiring cross-document synthesis. The integration of graph-based aggregation helps models maintain continuity across concepts, improving long-range context understanding.

Third, the framework improves **interpretability**, as each stage—retrieval, structuring, fusion, and generation—can be inspected independently. This transparency is valuable in domains where reasoning traces and evidence accountability are essential, such as healthcare, finance, legal systems, and academic research support.

Fourth, the modular architecture supports **domain adaptability**. Because the framework does not rely on domain-specific heuristics, it can be applied to diverse knowledge environments, including technical manuals, scientific literature, legal documents, and enterprise data systems.

Finally, the combination of retrieval precision, structured reasoning pathways, and graph-guided fusion contributes to **enhanced robustness**, allowing the system to produce more reliable answers even in ambiguous, incomplete, or noisy information settings. These advantages collectively position the framework as a meaningful step toward developing future LLM systems that are more trustworthy, evidence-aware, and aligned with real-world reasoning demands.

CHAPTER 6

EXPERIMENTAL VALIDATION

6.1 Overview

This chapter presents an empirical evaluation of a two-stage retrieval pipeline designed to improve passage selection in a Retrieval-Augmented Generation (RAG) setting. The goal of the experiment is to measure how much a simple reranking stage, added on top of dense retrieval, can improve retrieval quality for conversational question answering. Rather than implementing a full Graph-RAG system, this study focuses on a practical, lightweight architecture that aligns with the thesis theme of improving retrieval quality and context integration.

The evaluation uses a subset of the CoQA dataset, which contains multi-turn conversational questions paired with supporting passages. Retrieval quality is measured using Hit@1 and Exact Match at 1 (EM@1), computed for a baseline dense retriever and a two-stage (dense + reranker) pipeline.

6.2 Dataset and Experimental Setup

6.2.1 CoQA Dataset

CoQA (Conversational Question Answering) is a large-scale benchmark containing conversational QA pairs grounded in short passages. Each example consists of:

- A context passage (story).
- A sequence of questions asked about the passage.
- A corresponding sequence of answers.

For this thesis, only the text portion of CoQA is used. The goal is not to generate answers, but to evaluate how well the retrieval pipeline can identify the passage that contains the gold answer.

A preprocessed table is constructed with the following fields for each QA pair:

- split (train or dev)
- story_id
- context (passage text)
- question_id
- question
- answer

From the full CoQA corpus, a random subset of 500 QA pairs is sampled to form a working dataset. This subset is large enough to include diverse stories and question types, while remaining small enough to support iterative development.

6.2.2 Evaluation Subset

The evaluation subset consists of 100 CoQA questions that:

- Cover different stories and topics.
- Include a mixture of factual, descriptive, and light reasoning questions.

For each question:

- The **input** to the retrieval system is the question text.
- The **ground truth** consists of the gold answer string and its associated CoQA passage.

To ensure a controlled comparison, both the baseline and two-stage pipelines are evaluated on **exactly the same 100 questions** with identical gold labels. The only variable that changes between conditions is the retrieval strategy.

6.2.3 Embedding Models and Implementation

Two pretrained models are used without any task-specific fine-tuning:

- Dense Retriever: all-MiniLM-L6-v2 sentence embedding model, used as a bi-encoder to embed questions and passages into a shared vector space.
- Reranker: cross-encoder/ms-marco-MiniLM-L-6-v2, used to jointly score (question, passage) pairs via cross-attention.

All experiments are implemented in Python using standard libraries for embedding, similarity computation, and metrics. The pipeline follows these steps:

1. Precompute dense embeddings for all 437 context passages once.
2. For each of the 100 evaluation questions, encode the question and perform dense retrieval.
3. Optionally apply CrossEncoder reranking over the top-k candidates.
4. Compute evaluation metrics based on the final Top-1 retrieved passage.

6.3 Retrieval Pipelines

6.3.1 Baseline: Dense Retrieval Only

The baseline pipeline is a single-stage dense retriever:

1. **Question Encoding**

   The question is encoded using all-MiniLM-L6-v2 into a fixed-dimensional vector.

2. **Similarity Computation**

   Cosine similarity is computed between the question vector and the embedding of each of the 437 passages.

3. **Top-1 Selection**

   Passages are ranked by similarity score, and the **single most similar passage** is selected as the Top-1 retrieved context.

4. **Answer Alignment**

   For analysis, the pipeline records:

- Whether the Top-1 passage contains the gold answer string.
- Whether the gold answer associated with the Top-1 passage matches the CoQA gold answer exactly.

This baseline represents a standard dense-retrieval-only RAG setup, where retrieval decisions depend solely on embedding similarity.

6.3.2 Two-Stage Pipeline: Dense Retrieval + CrossEncoder Reranking

The proposed pipeline adds a reranking stage on top of the dense retriever:

1. **Dense Candidate Retrieval (Top-5)**

   The dense retriever computes cosine similarity between the question embedding and all passage embeddings and retrieves the **Top-5** most similar passages. This serves as a fast, recall-oriented candidate generation step.

2. **CrossEncoder Reranking**

   The CrossEncoder reranker scores each (question, passage) pair among the Top-5 candidates. Because the CrossEncoder jointly attends to the question and passage tokens, it can model finer interactions than cosine similarity alone.

3. **Final Top-1 Selection**

   The Top-5 candidates are sorted by the CrossEncoder score, and the highest-scoring passage is chosen as the final Top-1 retrieved context.

4. **Answer Alignment**

   As with the baseline, the pipeline checks whether:

- The final Top-1 passage contains the gold answer string.

- The gold answer associated with the retrieved passage matches the CoQA gold answer exactly.

This two-stage design mirrors practical RAG systems that use dense retrieval for speed and a stronger reranker for precision.

6.4 Evaluation Metrics and Procedures

6.4.1 Hit@1

Hit@1 measures whether the Top-1 retrieved passage contains the correct answer string somewhere in its text.

1. For each question–answer pair:
2. Run the retrieval pipeline to obtain the Top-1 passage.
3. Convert both the gold answer and the retrieved passage to lowercase strings.

If the gold answer substring appears in the passage text, the trial is counted as a "hit"; otherwise, it is a miss.

Hit@1 is the fraction of hits across the 100 questions:

$$\text{Hit@1} = \frac{\text{Number of questions where answer appears in Top-1 passage}}{\mathbf{100}}$$

This metric emphasizes coverage: it checks whether the system retrieved a passage that even mentions the answer.

6.4.2 EM@1 (Exact Match at 1)

EM@1 is a stricter metric that checks whether the retrieved Top-1 passage is exactly the same passage that the dataset marks as the gold context.

For each question–answer pair:

Run the retrieval pipeline to obtain the Top-1 passage.

Compare the retrieved passage text with the gold context passage text.

If they match (after simple normalization such as trimming and lowercasing), the trial is counted as an exact match.

EM@1 is the fraction of exact matches across the 100 questions:

$$\text{EM@1} = \frac{\text{Number of questions where Top-1 passage equals gold passage}}{\mathbf{100}}$$

While Hit@1 allows any passage that contains the answer, EM@1 checks whether the system retrieves precisely the intended supporting passage.

6.4.3 Precision, Recall, and F1 Score

To provide classification-style metrics, the retrieval task is cast as a binary decision per question: **did the system successfully retrieve a passage containing the gold answer?**

For each pipeline:

- **True label (per question)**: 1, because for each question there exists at least one passage in the corpus that contains the correct answer.
- **Predicted label (per question)**: 1 if Hit@1 is a success for that question, 0 otherwise.

Using these labels:

- **Precision**: fraction of retrieved positives that are correct.
- **Recall**: fraction of questions for which the system successfully retrieves a correct passage (numerically equal to Hit@1 under this definition).
- **F1 score**: harmonic mean of precision and recall.

This view allows the use of familiar IR metrics (precision, recall, F1) directly on the retrieval outcomes.

6.4.4 Bootstrap Resampling

To assess robustness, a simple **bootstrap resampling** procedure is applied over the 100-question set:

1. Perform **30 bootstrap iterations**.
2. In each iteration, sample 100 questions **with replacement** from the original 100-question set.
3. Recompute Hit@1, EM@1, precision, recall, and F1 for both the baseline and two-stage pipelines on the resampled set.
4. Aggregate the metrics across iterations to obtain mean and standard deviation estimates.

This procedure provides a sanity check that the observed improvements are not tied to a specific fixed split of questions and that the metrics are relatively stable across different subsets.

6.5 Quantitative Results

6.5.1 Overall Metrics on 100 Questions

Table 3 reports the main retrieval metrics for the baseline dense retriever and the two-stage (dense + CrossEncoder) pipeline on the 100-question evaluation set.

Table 6.1 – Retrieval Performance on CoQA (100 Questions)

Metric	Baseline (Dense)	Two-Stage (Dense + Reranker)	Improvement (absolute / relative)
Hit@1	0.270	0.360	+0.090 (≈ +33%)
EM@1	0.250	0.310	+0.060 (≈ +24%)
Precision	1.000	1.000	0.000 (no change)
Recall	0.270	0.360	+0.090 (≈ +33%)
F1 score	0.425	0.529	+0.104 (≈ +24%)

Table – 3 Results (Performance Metrics)

Key observations:

- Hit@1 increases from 0.27 to 0.36, meaning the reranked pipeline retrieves a passage containing the gold answer for roughly 36 out of 100 questions, compared to 27 for dense retrieval alone.
- EM@1 improves from 0.25 to 0.31, indicating more exact alignment with the gold answer context.
- Precision remains at 1.0 for both systems under the adopted definition, reflecting that whenever the system predicts a success, the retrieved passage truly contains the gold answer.
- Recall and F1 both increase, highlighting that the gains are driven by retrieving correct passages more often, not by expanding the set of predicted positives.

6.5.2 Bootstrap Analysis

Across 30 bootstrap resamples, the metrics for both pipelines remain stable, with small standard deviations around the means. The two-stage pipeline consistently outperforms the baseline on Hit@1, EM@1, recall, and F1 in nearly all resamples.This indicates that the observed improvements are not artifacts of a specific subset of questions. Instead, the reranking stage provides a robust, repeatable benefit to retrieval quality across slightly different realizations of the evaluation set.

6.6 Qualitative Analysis

Beyond aggregate metrics, several individual examples illustrate the effect of reranking:

- In cases where multiple passages share overlapping vocabulary with the question, the **dense retriever** often selects a passage that is topically related but does not contain the exact answer span.
- The **CrossEncoder reranker** can down-rank such false positives when the fine-grained token-level interaction between question and passage reveals a mismatch, and promote passages whose content better aligns with the answer.

Typical qualitative patterns include:

- Questions requiring **specific details** (names, numbers, short facts) benefit particularly from reranking, as the CrossEncoder is more sensitive to exact lexical and semantic matches.
- For **broader or paraphrased questions**, the dense retriever sometimes already selects a suitable passage, and the reranker tends to preserve this choice, indicating that it does not degrade good candidates.

These examples support the quantitative findings by showing how reranking corrects dense retrieval errors at the instance level.

6.7 Discussion

The experiments demonstrate that even a relatively simple two-stage retrieval architecture—dense Top-k retrieval followed by CrossEncoder reranking—can provide **measurable improvements** in retrieval quality in a RAG pipeline.

- Because the generation component is held fixed, the results isolate the impact of **retrieval architecture** on grounding quality, which aligns with the thesis emphasis on retrieval design.
- The bootstrap analysis, while lightweight, adds credibility by showing that the effect is consistent across resampled subsets, not just a single fixed split.

At the same time, there are clear limitations:

- The current setup still treats retrieval as a **single-shot** operation per question; more advanced designs such as dynamic or agentic RAG could further refine retrieval over multiple steps.
- Only Top-1 retrieval is evaluated; analyzing Top-k recall and downstream answer generation quality would give a more complete picture of end-to-end performance.

Overall, the experimental validation supports the central claim of the thesis: improving retrieval—specifically through a multi-stage pipeline with reranking—meaningfully enhances the grounding quality available to RAG systems and provides a practical bridge toward richer, more structured retrieval architectures.

6.8 Summary

This chapter presented an empirical evaluation of a two-stage retrieval pipeline on 100 CoQA questions. Using MiniLM for dense retrieval and a CrossEncoder for reranking, the system improved Hit@1 from 0.27 to 0.36 and EM@1 from 0.25 to 0.31 relative to a dense-only baseline.

CHAPTER 7

STATE-OF-THE-ART

7.1 Overview of State-of-the-Art RAG Systems

Recent advancements in Retrieval-Augmented Generation (RAG) reflect a shift from simple retrieval-plus-generation pipelines toward structured, reasoning-aware, and graph-integrated architectures. Traditional RAG models rely on vector-based retrieval and linear evidence fusion, whereas state-of-the-art systems increasingly incorporate multi-step reasoning, entity-level structuring, and graph-based relational modeling to reduce hallucinations and improve factual grounding. These developments represent a broader transition from "retrieval-focused augmentation" to "reasoning-centric augmentation," where the goal is no longer just to provide more context, but to integrate that context in a way that supports deeper, interpretable inference paths.

Modern systems such as **GraphRAG**, **Enhanced GraphRAG**, **CoT-RAG**, and **NotebookLM** push the boundaries of how LLMs can leverage external knowledge. They combine dense retrieval, knowledge graph construction, relational clustering, and chain-of-thought alignment to enable models to reason over structured evidence instead of isolated text chunks. These approaches improve relevance, reduce redundancy, and unlock multi-hop reasoning capabilities that traditional RAG pipelines lack.

State-of-the-art implementations also emphasize dynamic retrieval timing (e.g., FLARE, ReAct-style prompting), hybrid retrievers, and graph-guided aggregation strategies that ensure retrieved evidence is not merely appended to a prompt but fused into a coherent, explanation-ready structure. This evolution enables LLMs to provide more consistent, interpretable, and domain-aware outputs, especially for complex, multi-document or multi-entity tasks.

7.2 GraphRAG and Enhanced GraphRAG

GraphRAG represents a significant advancement over traditional Retrieval-Augmented Generation by shifting from **flat text retrieval** to **structured, graph-based knowledge retrieval**. Instead of treating retrieved chunks as isolated pieces of information, GraphRAG organizes them as **nodes** and **edges**, capturing relationships, hierarchies, and semantic dependencies across the corpus. This allows LLMs to perform **multi-hop reasoning**, follow explicit relational paths, and retrieve evidence that is structurally relevant rather than merely textually similar.

GraphRAG pipelines typically involve:

1. **Entity and Relationship Extraction** – converting raw text into graph representations.
2. **Graph Construction** – forming local subgraphs or global knowledge graphs.
3. **Graph-Based Retrieval** – retrieving not only relevant nodes but their connected neighbors to preserve contextual meaning.
4. **Graph-Guided Prompting** – supplying the LLM with retrieved graph clusters, relationship paths, and structured summaries.

This graph-oriented approach improves:

- **Contextual coherence**, since retrieved information is semantically linked.
- **Interpretability**, because reasoning paths (A → B → C) can be exposed.
- **Evidence diversity**, through graph expansion and neighborhood exploration.
- **Multi-hop reasoning**, which traditional RAG struggles with.

Enhanced GraphRAG extends this paradigm by introducing:

- **Cluster-based summarization** of large corpora into topic-specific subgraphs.
- **Weighted edges** based on semantic strength or frequency.

- **Graph propagation techniques** to expand retrieval depth adaptively.
- **Hybrid graph + dense retrieval** to balance structural precision and recall.

These improvements make Enhanced GraphRAG better suited for **large, complex knowledge bases**, providing higher recall, more stable reasoning, and reduced hallucination rates. The combination of graph-structured retrieval and reasoning-aware generation positions GraphRAG as a foundational step toward future context-aware LLM systems capable of scalable, interpretable, and relational reasoning.

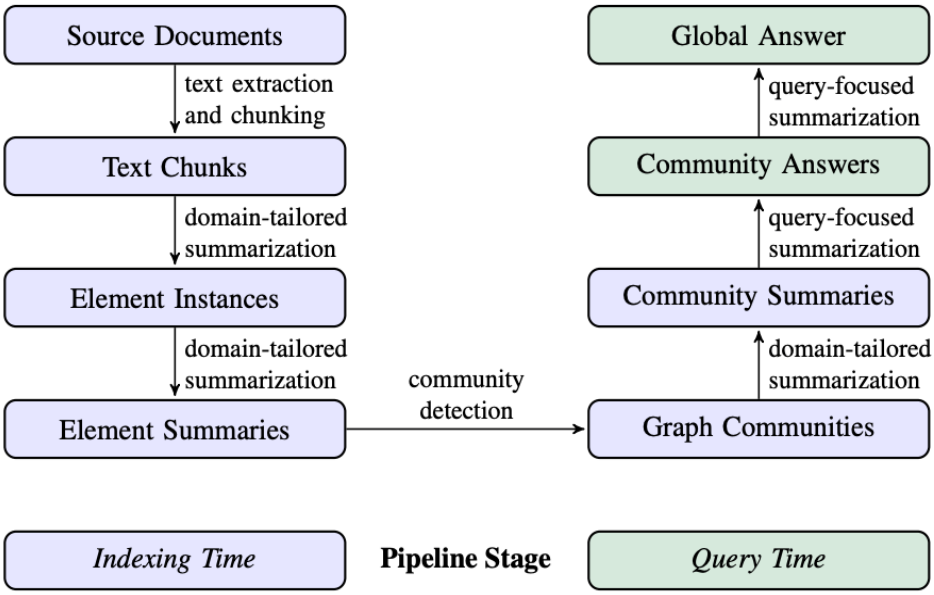

Figure 3: Graph RAG pipeline

7.3 NotebookLM: A Real-World Application of Retrieval-Augmented Generation

NotebookLM, developed by Google Labs, represents one of the most mature and practical implementations of Retrieval-Augmented Generation (RAG) deployed for real-world knowledge work. Unlike controlled research prototypes, NotebookLM is designed for everyday users—students, researchers, analysts, and content creators—who routinely need to work with large collections of documents. Its design demonstrates how RAG systems can move beyond

experimental settings and become reliable tools for interactive reasoning, summarization, and context-driven analysis.

At its core, NotebookLM enables users to upload diverse information sources—including PDFs, Google Docs, slides, and web pages—and then engage in conversational queries grounded directly in those materials. The system retrieves relevant segments from user-provided content, synthesizes them into contextual responses, and supports multi-source reasoning. This real-world deployment highlights two significant strengths of RAG: factual grounding and adaptability to custom datasets. By restricting the retrieval space to only the documents provided by the user, NotebookLM minimizes hallucinations and ensures that generated responses remain anchored to verifiable content.

NotebookLM incorporates several advanced features that showcase the evolution of RAG beyond simple retrieval-and-generate loops. Its Mind Map functionality visualizes relationships between concepts extracted from the uploaded sources, illustrating how structured or graph-like representations can enhance context comprehension. Similarly, its Audio Overviews leverage long-form summarization to provide digestible narrative explanations, demonstrating the usefulness of multimodal generative capabilities layered on top of RAG pipelines. These features collectively emphasize that effective RAG systems not only retrieve information, but also help users interpret, organize, and navigate complex knowledge spaces.

As a state-of-the-art consumer-facing system, NotebookLM provides an important benchmark for evaluating the practical reliability, interpretability, and usability of RAG frameworks. It underscores the increasing demand for tools that combine retrieval precision with rich, context-aware reasoning, and serves as an operational example of how future RAG and GraphRAG models may be deployed at scale in real-world environments.

7.3.1NotebookLM Architecture Overview

At a high level, NotebookLM follows a structured RAG pipeline consisting of:

1. **Ingestion and Normalization**

   User-provided sources are parsed and converted into normalized textual representations, preserving lightweight structure such as sections, headings, and timestamps.

2. **Semantic Chunking and Representation**

   Documents are segmented into semantically coherent chunks rather than fixed-length windows. These chunks are embedded into a dense vector space using proprietary encoders and stored in an indexed retrieval system along with metadata.

3. **Query Analysis and Rewriting**

   A Gemini-family model analyzes the user query to determine intent, scope, and constraints. The system may rewrite or expand the query to improve retrieval effectiveness while separating task framing from evidence requirements.

4. **Hybrid Retrieval and Reranking**

   NotebookLM performs dense vector retrieval augmented with lexical and metadata-based filtering. A reranking stage reorders top candidates to reduce redundancy and improve relevance.

5. **Context Grounding and Generation**

   Retrieved chunks are assembled into a grounded context bundle, and Gemini is instructed to generate answers strictly from this context with explicit source citations.

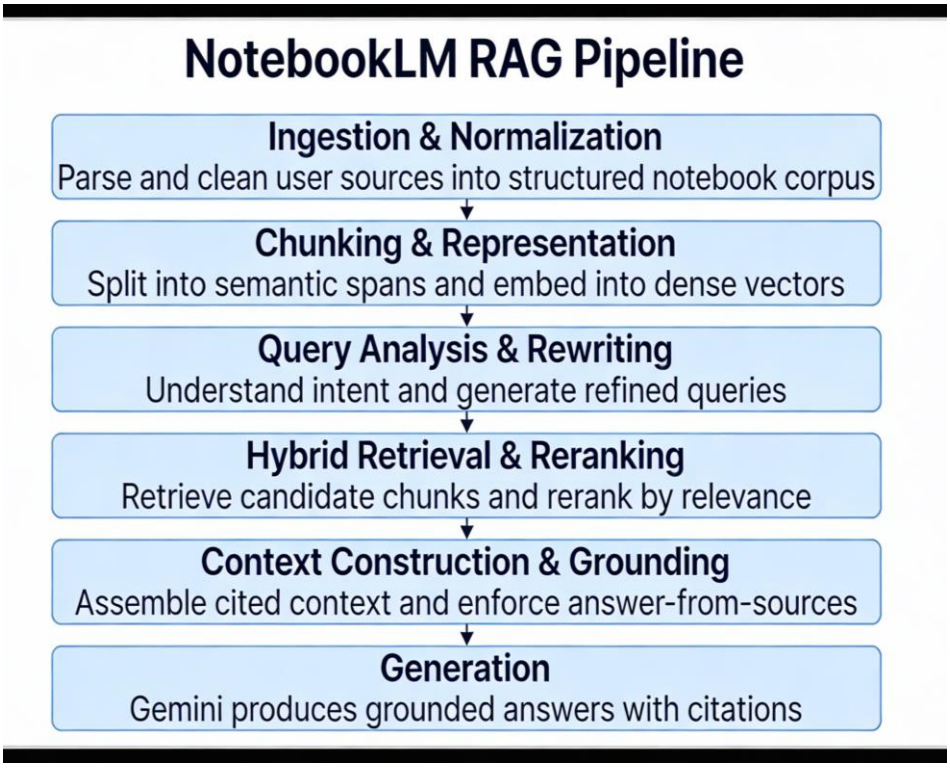

Figure 4 : NotebookLM RAG pipeline architecture

A defining characteristic of NotebookLM is its emphasis on **answer-from-sources behavior**, enforcing citation-based grounding more strictly than general-purpose RAG systems.

7.4 Limitations of NotebookLM and Static RAG Systems

Despite its robustness, NotebookLM relies on a largely static retrieval pipeline. Query analysis, retrieval, and reranking are typically executed as a single pass per user request, without iterative refinement based on intermediate outcomes.

Key limitations include:

- Lack of explicit iterative retrieval loops
- Limited adaptability to query complexity
- Fixed retrieval strategies regardless of task difficulty
- Absence of explicit decision-making or planning components
- Restricted architectural transparency due to closed-system design

These constraints can hinder performance on tasks requiring multi-step reasoning, hypothesis refinement, or exploratory information synthesis.

7.5 Beyond Standard RAG: Agentic RAG (Future Extension)

Agentic RAG extends standard RAG architectures by introducing an agent or controller that can make decisions during the retrieval process. Instead of executing a fixed retrieval–generation pipeline, an agentic system dynamically determines how many retrieval steps are needed, how queries should be refined, and which tools or data sources should be used.

Core capabilities of Agentic RAG include:

- Iterative query refinement based on intermediate results
- Dynamic tool selection (retrievers, filters, external APIs)
- Multi-step reasoning across multiple retrieval cycles
- Adaptive retrieval strategies based on task complexity

7.6 Challenges in Evaluating GraphRAG Systems

Evaluating GraphRAG systems presents several fundamental challenges due to the complexity of graph-structured retrieval, heterogeneous reasoning pathways, and the lack of standardized benchmarks. Unlike traditional RAG pipelines—where evaluation can rely on relatively simple metrics such as retrieval accuracy or answer relevance—GraphRAG systems introduce multi-hop paths, relational structures, and context fusion dynamics that require deeper and more nuanced evaluation methodologies.

First, **measuring multi-hop reasoning quality** remains difficult because evidence is not retrieved as flat text chunks but as interconnected subgraphs. Existing QA datasets often assume single-hop or loosely connected contexts, making them insufficient for evaluating graph-guided reasoning. Multi-hop datasets (e.g., HotpotQA) partially address this gap but still do not capture

the full granularity of entity-relationship graphs used in GraphRAG.

Second, **evaluating graph construction quality** is challenging due to variability in entity extraction, relationship inference, and clustering. Errors in any of these steps compound downstream, making it difficult to distinguish whether performance degradation is caused by retrieval, structuring, or generation. Human evaluation of graph correctness is costly and inconsistent, limiting scalability.

Third, **context fusion evaluation lacks standardized metrics**. Graph-aware fusion aims to combine relational evidence more coherently than simple concatenation, but there is no established metric for measuring coherence across structured evidence. Metrics such as BLEU or ROUGE fail to capture relational correctness, while embedding-based similarity overlooks graph topology.

Fourth, **interpretability and traceability challenges** complicate evaluation. Although GraphRAG is designed to improve transparency via reasoning paths, quantifying the usefulness or correctness of these paths requires human judgment. Automated scoring systems are still underdeveloped, especially for verifying multi-hop relational chains.

Finally, **computational cost and scalability** affect evaluation feasibility. GraphRAG pipelines—especially those with dynamic graph updates or dense entity-relation extraction—can be resource-intensive, limiting large-scale experimentation. As a result, evaluation results often depend heavily on hardware, dataset size, and implementation choices, reducing comparability across studies.

Overall, the evaluation of GraphRAG systems remains an open challenge, requiring new benchmarks, graph-aware metrics, and improved automated evaluation tools to accurately reflect the strengths and limitations of graph-structured retrieval and reasoning.

CHAPTER 8

DISCUSSION AND IMPLICATIONS

8.1 Key Findings

This thesis investigated how retrieval strategies within Retrieval-Augmented Generation (RAG) pipelines influence the grounding and reliability of Large Language Model (LLM) outputs. By combining a focused literature survey with controlled experimental evaluation, several key findings emerged.

1. **Retrieval quality is a primary determinant of grounding accuracy in RAG systems.**

   Across both literature and experimental analysis, retrieval precision was found to play a dominant role in determining the factual correctness of generated responses. Even highly capable generation models fail when retrieval introduces noisy or weakly relevant context.

2. **Single-stage dense retrieval provides reasonable recall but introduces irrelevant context.**

   Bi-encoder-based dense retrieval improves semantic matching compared to purely lexical methods; however, it often retrieves passages that are topically related but insufficiently precise, reducing grounding quality.

3. **Two-stage retrieval with cross-encoder reranking significantly improves retrieval accuracy.**

   Experimental results demonstrate that incorporating a cross-encoder reranking stage leads to consistent improvements in Hit@1 and Exact Match@1 (EM@1), confirming that reranking effectively filters noisy dense retrieval outputs.

4. **Retrieval architecture improvements alone can yield measurable gains without model retraining.**

The observed improvements were achieved using pretrained models without fine-tuning, highlighting that architectural choices in retrieval pipelines can substantially improve performance even under constrained settings.

5. **Static retrieval pipelines limit adaptability to complex information needs.**

   Both literature review and empirical findings indicate that fixed, single-pass retrieval strategies struggle with ambiguity, multi-step reasoning, and exploratory queries, motivating more adaptive retrieval mechanisms.

6. **Real-world RAG systems emphasize multi-stage retrieval and grounding constraints.**

   Systems such as NotebookLM reflect industry best practices by combining semantic chunking, hybrid retrieval, reranking, and strict source grounding, aligning closely with the experimental insights presented in this thesis.

8.2 Conclusion

This thesis set out to examine how retrieval strategies influence the effectiveness of Retrieval-Augmented Generation systems in grounding Large Language Model outputs. While RAG mitigates hallucinations by incorporating external context, its success depends critically on the precision and structure of the retrieval pipeline.

Through a review of existing RAG literature, this work highlighted a clear trend toward multi-stage retrieval architectures that prioritize grounding and relevance. Practical systems increasingly rely on semantic chunking, reranking, and constrained generation to improve reliability, yet many pipelines remain static and limited in adaptability.

The experimental component of this thesis provided a controlled comparison between single-stage dense retrieval and two-stage retrieval incorporating cross-encoder reranking. Results on the CoQA dataset showed that two-stage retrieval consistently outperforms dense

retrieval alone, as measured by Hit@1 and EM@1. These findings demonstrate that improving retrieval architecture—without modifying the generation model—can yield meaningful gains in grounding accuracy.

Overall, this work reinforces the conclusion that retrieval quality is a central bottleneck in RAG systems. By empirically validating the effectiveness of reranking-based retrieval pipelines and contextualizing these results within real-world RAG deployments, this thesis contributes to a clearer understanding of how retrieval design choices impact the reliability of LLM-based systems.

8.3 Future Research Directions

While this thesis focuses on static retrieval pipelines, several promising directions remain for future research.

1. **Agentic Retrieval-Augmented Generation**

   Future work may explore Agentic RAG systems in which a controller model dynamically refines queries, selects retrieval tools, and determines when additional retrieval steps are necessary. Such approaches could overcome the limitations of single-pass retrieval.

2. **Iterative and Multi-Hop Retrieval Evaluation**

   Extending evaluation to datasets requiring explicit multi-hop reasoning would provide deeper insight into how retrieval pipelines perform under complex reasoning demands.

3. **Hybrid Retrieval Strategies**

   Combining dense, sparse, and metadata-based retrieval methods in adaptive ways may further improve retrieval robustness across domains and query types.

4. **Retrieval-Aware Generation Evaluation**

   Future studies could investigate how retrieval improvements propagate to downstream

generation quality using generation-focused metrics alongside retrieval metrics.

5. **Scalability and Real-World Deployment**

   Evaluating retrieval pipelines under larger corpora, longer contexts, and real-time constraints would enhance the applicability of RAG systems in production environments.

6. **Multimodal Retrieval-Augmented Generation**

   Extending retrieval pipelines to incorporate non-textual modalities such as images, tables, and structured data represents an important avenue for future RAG research.

7. **Safety, Transparency, and Provenance**

   Future work should integrate uncertainty estimation, citation verification, and provenance tracking to support trustworthy deployment of RAG systems in high-stakes applications.

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